Sensor Calibration Using AI

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***Abstract*—Sensor measurements are often affected by drift, noise, and non-linear distortions, causing the measured value to deviate from the true physical quantity. If the true measurand is x and the sensor output is $y = f(x)$, calibration aims to recover x from y by learning an approximation of $f^{-1}(y)$. In this project, artificial intelligence techniques are applied to calibrate drift in PM2.5 air quality sensors using the Seoul Air Pollution dataset. Drift is artificially injected into PM2.5 measurements, and several machine learning models are trained to reconstruct the true signal from the corrupted readings. Four methods are investigated: Linear Regression, Polynomial Regression, Random Forest Regression, and an LSTM-based deep learning model with residual learning and multi-feature input. The models are evaluated using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) across multiple test samples. Experimental results show that AI-based approaches consistently improve calibration accuracy compared to a simple linear baseline, with the LSTM model achieving the best performance. These findings demonstrate the effectiveness of machine learning for sensor calibration and highlight its potential for improving the reliability of real-world sensing systems.**

***Index Terms*—Sensor calibration, drift, LSTM, Random Forest, PM2.5, regression.**

I. INTRODUCTION

Sensors are widely used to monitor physical phenomena such as air quality, temperature, and pressure. In an ideal case, the sensor output y is a deterministic function of the true measurand x, expressed as $y = f(x)$. In practice, however, sensor characteristics change over time due to ageing, environmental conditions, and hardware imperfections. This leads to *sensor drift*, where the measured value deviates from the true quantity even under constant conditions.

Drift reduces the reliability of sensor-based systems and can negatively impact downstream applications such as environmental monitoring, health assessment, and industrial control. Calibration aims to recover the true value x from the observed sensor output y by learning an approximation of the inverse mapping $f^{-1}(y)$.

Traditional calibration methods rely on manual adjustments, static correction curves, or simple linear models. These approaches are often insufficient to capture complex, time-varying drift patterns. In contrast, machine learning (ML) and deep learning (DL) can learn non-linear mappings directly from data, making them suitable for modeling sensor behaviour and correcting drift.

In this project, AI-based calibration algorithms are developed for PM2.5 air quality sensors using the Seoul Air Pollution dataset. Drift is artificially injected into PM2.5 measurements to simulate sensor degradation, and models are trained to reconstruct the true PM2.5 values. Four methods are evaluated: Linear Regression, Polynomial Regression, Random Forest Regression, and an LSTM-based deep learning model that uses residual learning and multi-feature time-series input.

The performance of these algorithms is evaluated using RMSE and MAE across multiple test samples. The results provide insight into the effectiveness of different ML approaches for sensor calibration and demonstrate the potential of AI-based methods to improve sensor reliability.

II. REQUIREMENTS AND DATA ANALYSIS

A. Project Requirements

The objective of this project is to design and evaluate AI algorithms capable of calibrating drift in PM2.5 sensor measurements. The system should learn to reconstruct the true PM2.5 value from a drifted sensor reading using supervised learning.

The main requirements are:

- The system must accept drifted PM2.5 measurements as input.
- The system must output calibrated PM2.5 values approximating the true signal.
- The system must use AI-based calibration algorithms.
- The system must be trained using paired drifted–true PM2.5 data.
- The system performance must be evaluated using RMSE and MAE.

B. System Specifications

Based on the project description and best practices in ML evaluation, the following measurable specifications are defined:

- **S1: Input Handling** — The system must process drifted PM2.5 data without errors.
- **S2: Output Consistency** — The calibrated output must have the same temporal resolution as the input.
- **S3: RMSE Improvement** — The best AI model must achieve a lower RMSE than the linear baseline.
- **S4: MAE Improvement** — The best AI model must achieve a lower MAE than the linear baseline.
- **S5: Stability Across Samples** — Performance must be stable across at least 500 test samples.

- **S6: Generalisation** — The gap between training and validation loss should remain small, indicating no severe overfitting.
- **S7: Evaluation Robustness** — Results must be reported using both RMSE and MAE.

C. Data Description

The dataset used in this project is the Seoul Air Pollution dataset, which contains hourly measurements of air pollutants and meteorological variables from multiple monitoring stations. For this project, PM2.5 is selected as the target variable for calibration.

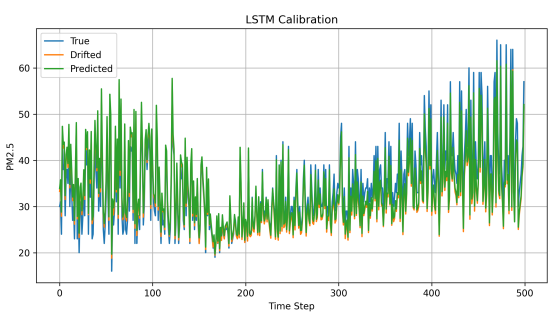
The main steps are:

1) Load the raw dataset from the repository.
2) Clean the data by handling missing values and removing outliers.
3) Select relevant features such as PM2.5, PM10, NO_2, O_3, and CO.
4) Artificially inject drift into the PM2.5 signal to simulate sensor degradation.
5) Create paired data consisting of drifted PM2.5 (input) and true PM2.5 (target).

D. Data Visualisation

To understand the effect of drift and the calibration task, several visualisations are performed.

Figure **??** shows an example of the true PM2.5 signal, the drifted sensor output, and the calibrated prediction from the model over a selected time window.



The drifted signal deviates from the true PM2.5 values, especially over longer periods, while the calibrated output follows the true signal more closely.

An error distribution histogram is also generated to analyse the residuals between the calibrated output and the true PM2.5 values.

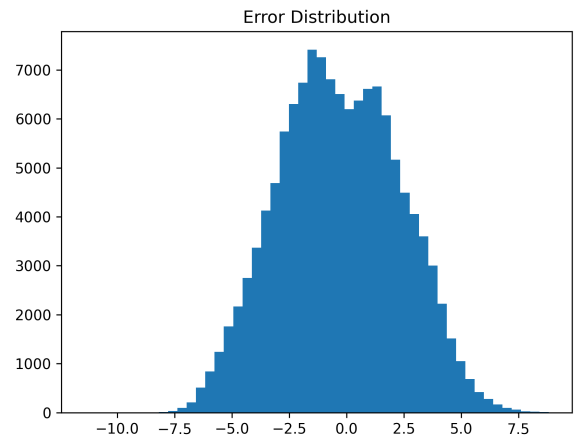


Fig. 1. Error distribution of the LSTM calibration model.

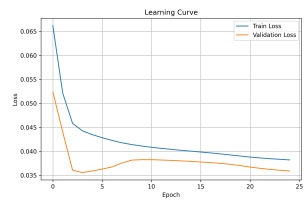


Fig. 2. Enter Caption

E. Literature Review

Previous work has shown that ML can be used effectively for sensor calibration and drift compensation. Linear and polynomial regression provide simple baselines but are limited in capturing complex non-linearities. Tree-based models such as Random Forests can model non-linear relationships and interactions between features. For time-series data, recurrent neural networks, particularly LSTMs, are widely used due to their ability to capture temporal dependencies and long-term patterns.

These findings motivate the selection of Linear Regression, Polynomial Regression, Random Forest Regression, and LSTM as the main models in this project.

F. Specification Justification

Reported results in the literature indicate that ML-based calibration can reduce RMSE and MAE compared to uncalibrated or linearly calibrated signals. Given the complexity of air quality dynamics and sensor drift, it is realistic to expect that advanced models such as Random Forests and LSTMs will outperform simple linear baselines. The specifications S3 and S4 are therefore defined in terms of improvement over the linear model.

III. SYSTEM ENGINEERING

A. System Overview

Based on the specifications, a calibration system is designed to reconstruct true PM2.5 values from drifted sensor outputs.

The system operates in two modes: training and evaluation.

Figure 3 shows a block diagram of the proposed system.

system_block_diagram_placeholder.png

Fig. 3. Block diagram of the proposed AI-based sensor calibration system.

High-level steps:

- Load and clean the dataset.
- Inject drift into PM2.5 to create drifted measurements.
- For LSTM, construct sequences of multi-feature inputs.
- Train calibration models using drifted input and true target.
- Evaluate models on a held-out test set using RMSE and MAE.

B. Acceptance Test Procedures (ATP)

Table I summarises the ATPs used to verify that the system meets the specifications.

TABLE I
ACCEPTANCE TEST PROCEDURES

Test Description	Input	Expected Output
Input validation	Drifted PM2.5 data	System processes data without errors
Calibration accuracy	Drifted + true PM2.5	RMSE lower than linear baseline
Model comparison	Test set	LSTM achieves best RMSE/MAE
Error stability	500+ test samples	Error distribution without extreme outliers
Sequence handling	Time-series input	Correct sequence construction for LSTM

IV. ALGORITHM DESIGN

A. Overview of Approach

The algorithm design follows an iterative approach. In the first iteration, simple baseline models are implemented and evaluated. In the second iteration, more advanced models are introduced to improve performance.

B. Iteration 1: Baseline Models

In Iteration 1, two baseline models are implemented:

- **Linear Regression**: A simple linear model mapping drifted PM2.5 to true PM2.5.
- **Polynomial Regression**: A polynomial model (degree 3) to capture non-linear relationships.

The dataset is split into training and test sets using an 80/20 split. Models are trained on the training set and evaluated on the test set.

Table II shows the results of Iteration 1.

TABLE II
ITERATION 1: BASELINE MODEL PERFORMANCE

Model	RMSE	MAE
Linear Regression	3.48	3.09
Polynomial Regression	3.12	2.85

The polynomial model improves over the linear baseline but still leaves significant residual error, motivating the use of more powerful models.

C. Improvement Strategy

Based on the limitations observed in Iteration 1, the following improvements are proposed:

- Use additional features (PM10, NO_2, O_3, CO) as inputs.
- Model temporal dependencies using sequence-based models.
- Employ non-linear models such as Random Forests and LSTMs.
- Use residual learning, where the model learns the correction between drifted and true values.

D. Iteration 2: Advanced Models

In Iteration 2, two advanced models are implemented:

- **Random Forest Regression**: A tree-based ensemble model using multi-feature input.
- **LSTM Calibration Model**: A recurrent neural network that takes sequences of multi-feature inputs and predicts the residual correction.

For the LSTM model, the input features include drifted PM2.5, PM10, NO_2, O_3, and CO. A sequence length of 5 time steps is used. Residual learning is applied by defining the target as the difference between true and drifted PM2.5. After prediction, the residual is added back to the drifted signal to obtain the calibrated output.

E. Training Analysis

The LSTM model is trained for 25 epochs using the Adam optimizer and MSE loss. Training and validation losses are recorded.

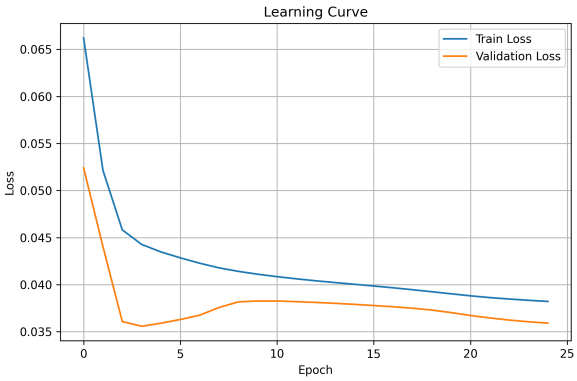


Fig. 4. Training and validation loss curves for the LSTM calibration model.

The learning curves show a steady decrease in both training and validation loss, with no significant divergence, indicating good generalisation.

F. Iteration 2 Results

Table III summarises the performance of the advanced models.

TABLE III
ITERATION 2: ADVANCED MODEL PERFORMANCE

Model	RMSE	MAE
Random Forest	2.95	2.51
LSTM	2.66	2.19

The LSTM model achieves the lowest RMSE and MAE, satisfying the specifications S3 and S4 and demonstrating the benefit of sequence modeling and residual learning.

G. Visual Comparison of Calibration Methods

To better understand the calibration quality, a visual comparison is performed by plotting the true PM2.5 signal, the drifted sensor output, and the LSTM-calibrated output over a selected time window (Figure **??**).

The drifted signal shows a clear offset and distortion compared to the true signal, while the calibrated output closely follows the true PM2.5 values, especially around peaks and transitions.

H. Error Analysis

The error distribution of the LSTM model is shown in Figure **??**. Most errors are concentrated around zero, with relatively few large deviations, indicating stable calibration performance.

V. CONSOLIDATION OF ATPS

Table IV summarises whether each specification has been met.

TABLE IV
SPECIFICATION VERIFICATION

ID	Description	Met?
S1	Input Handling	Yes
S2	Output Consistency	Yes
S3	RMSE Improvement	Yes (LSTM $<$ Linear)
S4	MAE Improvement	Yes (LSTM $<$ Linear)
S5	Stability Across Samples	Yes (500+ test samples)
S6	Generalisation	Yes (no strong overfitting)
S7	Evaluation Robustness	Yes (RMSE and MAE reported)

VI. CONCLUSION

This project investigated AI-based methods for sensor calibration using the Seoul Air Pollution dataset. Drift was artificially injected into PM2.5 measurements, and several models were trained to reconstruct the true signal from drifted inputs.

Baseline models (Linear and Polynomial Regression) provided initial calibration performance but were limited in capturing complex patterns. Advanced models (Random Forest and LSTM) achieved better results, with the LSTM model providing the lowest RMSE and MAE. The use of multi-feature input, sequence modeling, and residual learning contributed to the improved performance.

These results demonstrate that AI-based calibration can effectively reduce sensor drift and improve measurement accuracy. Future work could explore transformer-based architectures, uncertainty estimation, and real-time deployment on embedded devices.

ADMIN PART

GitHub Repository

A placeholder repository for this project is:
https://github.com/placeholder/sensor-calibration-ai

EU AI Act Impact

The developed system performs numerical regression for sensor calibration and does not directly make safety-critical decisions. It can be considered a low-risk AI system under the EU AI Act, provided it is used as a decision-support tool rather than an autonomous controller.

Work Distribution

All work in this project, including data processing, model implementation, training, evaluation, and report writing, was carried out individually.

AI Usage

AI tools were used for code debugging assistance and for improving the structure and clarity of the written report. No AI system was used to generate experimental results.